

[199]

SARDAR PATEL UNIVERSITY

B.C.A. [SEMESTER- 4] || [April -2026]

US04CBCA54: Operating Systems

Time: 02.00 pm – 05.00 pm



Date: 06-04-2026, Monday

Marks: 70

Q-1 Select the correct option from the following questions. (10)

- 1) Time sharing system is also known as _____.
a) Real-time b) Multi-tasking c) Multi-user d) None of these
- 2) A newly create process resides in _____ queue.
a) Job b) Ready c) Running d) Waiting
- 3) _____ scheduling algorithm gives minimal average waiting time.
a) FCFS b) SJF c) Round-robin d) Priority
- 4) Virtual Memory is commonly implemented by _____.
a) Fragmentation b) Compaction c) Demand Paging d) Backing store
- 5) Which of the following memory allocation strategy is fastest?
a) Best-fit b) Optimal-fit c) Worst-fit d) First-fit
- 6) Printer is an example of _____ resource.
a) Logical b) Physical c) Virtual d) None
- 7) Multiple process solution is also known as _____.
a) Paging Algorithm b) Baker's Algorithm
c) Bakery Algorithm d) None of these
- 8) Which of the following is NOT the valid resource utilization state?
a) Release b) Request c) Use d) Execution
- 9) _____ option of vi editor is used to delete a line.
a) dd b) yy c) cc d) None
- 10) _____ option of ls command will sort output according to size of file.
a) -S b) -c c) -b d) None

Q-2 Short Questions (Attempt any Ten) (20)

- 1) Define Operating system.
- 2) Write a brief note on Client-Server Model.
- 3) Define Long term scheduler.
- 4) What is Belady's Anomaly?
- 5) What is Compaction? For what purpose it is used?
- 6) Explain Best-fit memory allocation techniques.
- 7) Why is LINUX an open source?

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- 8) When Race conditions arise?
- 9) Explain any two necessary conditions of Critical section.
- 10) Explain cp command with all its attributes.
- 11) Explain mv command with all its attributes.
- 12) Explain who command in brief.

- Q-3** A Explain SJF scheduling algorithm in brief. (05)
 B Which are the functions performed by Operating System? (05)
- OR
- Q-3** A Explain Round-Robin scheduling algorithm with example. (05)
 B Explain process state and PCB in detail. (05)
- Q-4** A Define Page Fault. Explain page fault handling technique. (05)
 B Explain Paging techniques with advantages and disadvantages. (05)
- OR
- Q-4** A Explain Optimal Page replacement algorithm in detail. (05)
 B What is Fragmentation? List different types of Fragmentation. Explain in details. (05)
- Q-5** A What is Cooperative Process? Explain Producer-Consumer Problem. (05)
 B Explain algorithm 3 for two-process solution. (05)
- OR
- Q-5** A What is LINUX? Explain basic features of LINUX Operating System. (05)
 B What do you mean by Deadlock? Explain all necessary conditions for occurrence of deadlock. (05)
- Q-6** A Explain for loop in Linux. (05)
 B What is FAP? Explain various ways to change permission on a file. (05)
- OR
- Q-6** A Explain if and case statement in LINUX. (05)
 B Explain Environment variables of Linux. (05)